Manit Khera
Blackburn South, VIC 3130
0412 761 206
manitkhera26@gmail.com

Friday 18 September 2026

Dr Wannita Takerngsaksiri
Deakin Applied Artificial Intelligence Initiative
Deakin University
221 Burwood Highway
Burwood, VIC 3125

Dear Dr Takerngsaksiri,

RE: Application for the Benchmarking Agent Workflows for Software Development research internship

I am writing to apply for the Benchmarking Agent Workflows for Software Development research internship with the Deakin Applied Artificial Intelligence Initiative, supervised by yourself and Associate Professor Scott Barnett. I am a Bachelor of Software Engineering (Honours) student at Deakin, Burwood, with a Weighted Average Mark of 81.33 and 100 in both Backend Development and Data Structures and Algorithms. I am available on campus at Burwood two days per week for the full ten weeks, and I understand the position is unpaid.

As a child I wanted to be a scientist, which in practice meant finding experiments on YouTube, gathering whatever was in the house and trying them to see what changed. I found the same thing again in Year 12, in an IB Psychology Extended Essay I spent months on: one question, a body of literature to read and understand, and a conclusion I had to be willing to defend. I will not pretend I already know that research is the career for me. I do not. Being completely invested in one problem is one of the few things that makes me lose track of time, and ten weeks on a real question is how I would find out.

The objective I keep returning to is the second, identifying the coding tasks the benchmark runs on. On the Deakin Software Engineering Club committee I solely designed the defects seeded into a React application built as the teaching artefact for a debugging workshop, for around thirty attendees to find. Choosing the faults was harder and more interesting than building the application. A defect that is too obvious measures nothing. One that is too subtle measures only patience. The ones worth planting are those a reviewer will confidently declare fixed without having fixed them. A benchmark task set has that problem from the other direction: if every workflow passes every task the benchmark has told you nothing, and the tasks that earn their place are the ones where an agent can report a success it has not earned.

Your first expected outcome is a working harness with run logs for every workflow, and I have already had to take a position on what a log may be used for. The backend I built for my backend development unit records every dispatched and completed action to a queryable execution history, and the rollback engine reconstructs the robot's state from that record rather than from what the robot reports, because a half-applied reversal would otherwise log the robot at a position it was never at. Every reported action is replayed through simulation before anything is committed. A log is not yet evidence, and the work is deciding what the record must guarantee before anyone draws a conclusion from it. A comparison across workflows stands or falls on that.

Two gaps, named rather than left for week one. I use Claude Code daily in my own development, so I am a user of the thing this project studies, but I have not configured agents beyond ordinary use: no custom skills, no hooks, no worktree setups. I have also designed a measurement instrument without ever running a controlled experiment with formal controls and statistics. Your brief lists both as skills this internship develops, so I will not dress either up. Underneath them sits Python and C#, code review on a team repository as well as my own, and the habit of instrumenting a system so the record can answer questions afterwards. I would arrive having worked through the agent configuration surface in my own time.

My resume is attached, and every project in it links to public code and a working demonstration, so the claims can be checked rather than taken on trust. I would welcome the chance to discuss the project, and thank you for your consideration.

Yours sincerely,

Manit Khera